

4.3 Odkształcenie sprężyste

$$-\frac{d}{dx} \left(E(x) \frac{du(x)}{dx} \right) = 0$$

$$u(2) = 0$$

$$\frac{du(0)}{dx} + u(0) = 10$$

$$E(x) = \begin{cases} 3 & \text{dla } x \in [0, 1) \\ 5 & \text{dla } x \in [1, 2] \end{cases}$$

Gdzie u to poszukiwana funkcja
 $[0, 2] \ni x \rightarrow u(x) \in \mathbb{R}$

Wyprowadzenie sformułowania wariacyjnego

$$-\frac{d}{dx} \left(E(x) \frac{du(x)}{dx} \right) = 0 \quad \Omega = [0, 2]$$

$$E(x) = \begin{cases} 3 & \text{dla } x \in [0, 1) \\ 5 & \text{dla } x \in [1, 2] \end{cases}$$

$$u'(0) + u(0) = 10 \quad u(2) = 0$$

Prawostronny zerowy warunek brzegowy Dirichleta

$$V = \{v(x) : v(2) = 0\}$$

$$\int_0^2 -\frac{d}{dx} (E(x)u'(x)) v(x) dx = \int_0^2 0v(x) dx$$

$$-\int_0^2 (E'(x)u'(x) + E(x)u''(x)) v(x) dx = 0$$

$$\int_0^2 (0 \cdot u'(x) + E(x)u''(x)) v(x) dx = 0$$

$$\int_0^2 E(x)u''(x)v(x) dx = 0$$

$$[E(x)u'(x)v(x)]_0^2 - \int_0^2 E(x)u'(x)v'(x) dx = 0$$

$$E(2)u'(2)v(2) - E(0)u'(0)v(0) - \int_0^2 E(x)u'(x)v'(x) dx = 0$$

Z warunków brzegowych

$$E(2)u'(2) \cdot 0 - E(0)(10 - u(0))v(0) - \int_0^2 E(x)u'(x)v'(x) dx = 0$$

$$E(0)u(0)v(0) - \int_0^2 E(x)u'(x)v'(x) dx = 10E(0)v(0)$$

$$\int_0^2 E(x)u'(x)v'(x) dx - E(0)u(0)v(0) = -10E(0)v(0)$$

Ostatecznie

$$B(u, v) = \int_0^2 E(x)u'(x)v'(x) dx - E(0)u(0)v(0)$$
$$L(v) = -10E(0)v(0)$$